Jiachen Ye

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EDUCATION

Carnegie Mellon University, Pittsburgh, PA
Ph.D. in Civil and Environmental Engineering (SETS)
Aug 2025 - Present

- Ph.D. student in the Sustainable Energy and Transportation Systems (SETS) program.
- Coursework focused on Transportation Engineering and Machine Learning.

Tsinghua University, Beijing, China
B.S. in Civil Engineering (Transportation Engineering)
Sep 2020 - Jun 2025

- Ranked top 1 in class (GPA: 3.69/4.00).
- Completed coursework in Transportation Engineering (Traffic Analysis and Design, Road Engineering, Green Transportation Systems) and Computer Science/Machine Learning (Data Structures, Machine Learning, Artificial Intelligence).

RESEARCH INTERESTS

Dynamic Traffic Assignment, Differentiable Transportation Modeling, Deep Learning for Transportation Engineering.

PUBLICATIONS & CONFERENCE PRESENTATIONS

J. Ye, X. Ma, and S. Qian. "A link-based differentiable framework for predictive dynamic traffic equilibrium." *Transportation Research Board (TRB) 2027 Annual Meeting.* **Accepted for conference presentation**.

J. Ye, T. Azfar, and R. Ke. "Real-Time Transit O-D Data Collection via Edge-AI-based Passenger Re-Identification." *Transportation Research Board (TRB) Annual Meeting*, 2025.

J. Ye, D. Wang, S. Jia, X. Pei, Z. Yang, Y. Zhang, and S. C. Wong. "CVVLSNet: Vehicle Location and Speed Estimation Using Partial Connected Vehicle Trajectory Data." *IEEE Intelligent Transportation Systems Conference (ITSC)*, 2024.

RESEARCH EXPERIENCE

Differentiable Dynamic Traffic Assignment and Transportation System Calibration, Department of Civil and Environmental Engineering, Carnegie Mellon University
Aug 2025 - Present

- Investigating differentiable dynamic traffic assignment (DTA) and traffic equilibrium solvers to enable gradient-based optimization in transportation network models.
- Exploring the integration of dynamic origin-destination estimation (DODE), traffic assignment, and equilibrium computation into a unified computational graph for end-to-end transportation system calibration.

Research on Cross-terminal Multimodal Cooperative Sensing, Department of Automation, Tsinghua University
Nov 2023 - Apr 2025

- Collaborated with Dr. Pei's lab to investigate the use of connected vehicle (CV) data for improving traffic safety and efficiency.
- Developed a deep learning approach to estimate vehicle locations and speeds from partial CV trajectory data without relying on fixed roadside detectors.

- Proposed a representation of variable vehicle state information and explored extensions to larger road networks.

Real-Time Transit O-D Data Collection via Edge-AI-based Passenger Re-Identification, Department of Civil and Environmental Engineering, Rensselaer Polytechnic Institute
Jun 2024 - Sep 2024

- Collaborated with a public transit company as part of Dr. Ke's lab to develop an automated approach for transit origin-destination (O-D) data collection.
- Developed an edge-AI framework using NVIDIA Jetson AGX Orin for real-time passenger detection, tracking, and re-identification from onboard surveillance video.

VR Simulation for Fire Evacuation Studies, Department of Civil Engineering, Tsinghua University
Jul 2022 - Dec 2022

- Analyzed eye-tracking and electrodermal activity (EDA) data collected from VR-based fire evacuation experiments.
- Extracted and statistically analyzed physiological features, contributing to a study on age-related differences in physiological responses during fire evacuation.

VR/AR Human-Computer Interaction Technology, Department of Computer Science, Tsinghua University
Nov 2021 - Jun 2022

- Conducted literature review on human-computer interaction and cognitive load in collaboration with Dr. Shi's lab.
- Developed Unity code for real-time sensor-based leg posture detection, supporting applications in patient monitoring.

HONORS AND AWARDS

Tsinghua University "Passionate About Reading" Scholarship	2021
Silver Award, Tsinghua University Summer Social Practice	2021
Tsinghua University Academic Excellence Scholarship	2021, 2022
Tsinghua University Lu Liqian Scholarship	2022
Tsinghua University Research Innovation Excellence Scholarship	2022, 2023

SKILLS

Technical Skills:

- Programming: Python, C, C++, SQL
- Machine Learning: PyTorch
- Transportation and GIS: SUMO, QGIS
- Development Tools: Git

Languages:

- Chinese: Native
- English: IELTS (7.0), TOEFL iBT (96)